Chapter 5: The Classic IS-LM Model: The Most Influential Model in Macroeconomic History

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1 Overview

The IS-LM model is arguably the most influential macroeconomic model in economic history. Derived by John Hicks, a British economist and Nobel laureate, in 1937 in an article in Econometrica, it was intended to explain Keynes's *General Theory of Employment, Interest and Money*. It consists of the IS (investment-savings) and the LM (loan market) equations. For many intermediate-level macroeconomic classes, the IS-LM model is still the main framework for analysis over 80 years after its creation. Few models can claim such success. While the IS-LM model is fundamentally different from most modern macroeconomic models, it is useful to understand it and appreciate where it comes from.

Here I present the IS-LM model slightly differently to existing expositions, especially in terms of notation. I do this to make it comparable to the models we consider later on.

A fundamental difference between the IS-LM model and the modern approach is how the equations are justified. The modern approach is to derive each of the equations that define the equilibrium of the model by solving the decision problems faced by all agents, where their objectives and constraints are explicitly stated. In contrast, in the IS-LM model, the equations that define the equilibrium are loosely motivated behavioral relationships based on broad patterns in the data. Below I also briefly explain how the IS-LM model is connected to the Quantity Theory of Money, which was at the heart of Friedman and Schwartz's work that we briefly discuss.

2 The IS Equation

The starting point of the IS-LM model is an uncontroversial national accounting identity:

$$Y_t = C_t + I_t + G_t + X_t \tag{1}$$

where Y_t is output, C_t is private consumption, I_t is investment, G_t is government spending, and X_t is some exogenous spending that we do not explicitly include for our present purposes, such as net exports. All variables here are expressed in real terms as opposed to in nominal dollar value.

The term "IS equation", from investment-savings, originates from the observation that we can re-arrange this equation to yield

$$I_t = Y_t - C_t - G_t - X_t \tag{2}$$

Here the left-hand side represents aggregate investment, while the right-hand side represents aggregate savings. In the aggregate, the two have to be equal to one another. What translates accounting identity (1) into a model, is a theory of consumption, investment behavior and a theory of government spending and taxes. This is what we now turn to.

Before getting there, however, we write the accounting identity slightly differently than in (1). As I discussed in Chapter 2, an approach we follow is to translate non-linear equations to linear ones via (log) linear approximation around a steady state. While we can in principle state the IS-LM model directly in a linear form, it is useful to express it in deviation from a steady-state to connect more clearly with the models that follow.

Let us first define a steady state of the model as the solution of the model in the absence of any shocks. Denote the value of each variable in steady state by $\bar{Y}$,$\bar{I}$,$\bar{C}$,$\bar{X}$. Furthermore define $\hat{Y}_t = \frac{Y_t - \bar{Y}}{\bar{Y}}$, $\hat{C}_t = \frac{C_t - \bar{C}}{\bar{Y}}$, $\hat{I}_t = \frac{I_t - \bar{I}}{\bar{Y}}$, $\hat{X}_t = \frac{X_t - \bar{X}}{\bar{Y}}$. The national accounting identity (1) can now be written in terms of these re-defined variables as:

$$\hat{Y}_t = \hat{C}_t + \hat{I}_t + \hat{G}_t + \hat{X}_t \tag{3}$$

2.1 The Keynesian Consumption Function

Aggregate consumption is roughly proportional to output in the data. Motivated by this observation, Keynes conjectured that households consume a fixed fraction of their disposable income (i.e. income net of taxes). A simple formalization of this idea is

$$\hat{C}_t = \alpha(\hat{Y}_t - \hat{T}_t) + \xi_t^C \tag{4}$$

where $\hat{T}_t$ is taxes in deviation from steady state as fraction of steady state output and ξ_t^C is an exogenous disturbance. The coefficient α is between 0 and

1 and is called the marginal propensity to consume. For each additional dollar of disposable income, α of the dollar is consumed, and the remainder is saved. A reasonable value for α for the US, for example, is 0.65, computed by calculating the consumption to output ratio in the data.

2.2 The Keynesian Investment Function

Investment is the expenditure of corporations and firms on new machines, buildings and technologies to improve production. Included in investment is also sometimes residential housing. A key abstraction of the IS-LM model is to ignore the role investment has on the production capacities of the economy. Instead, the focus is on how investment influences aggregate spending via (1). The Keynesian investment function is

$$\hat{I}_t = -b\hat{r}_t + \xi_t^I \tag{5}$$

where $\hat{I}_t$ is investment at time t in deviation from steady state, b is a positive coefficient, and $\hat{r}_t$ is the *real rate of interest* at time t in percentage deviation from steady state defined as $\hat{r}_t = \frac{r_t - \bar{r}}{1+\bar{r}}$. In addition, there is an exogenous shock term, ξ_t^I, that can be used to model changes in business confidence or other shocks to investment. Keynes famously referred to such shocks as "animal spirits."

Investment is a decreasing function of the real interest rate. As interest rates fall, the Keynesian argument goes, it is cheaper (in terms of both absolute and opportunity cost) for companies to borrow and make investments in their own business, or alternatively, for people to invest in housing, as they face lower mortgage payments. As in the case of the Keynesian consumption function, the investment function is not derived, but conjectured on based on observed behavior.

2.3 Government Spending and Taxes

Government spending and taxes are typically modeled as exogenous in the IS-LM model. $\hat{G}_t$ is the government spending and is affected by government policies, such as austerity measures or, in studies of the Great Depression, the New Deal and the Second World War. Examples of variation in taxes include the Kennedy, Reagan, Bush, and Trump tax cuts.

2.4 The Investment Savings Equation (IS)

We now have all that we need in order to produce the IS equation. We can plug in equation (4) and (5) into (3) for $\hat{C}_t$ and $\hat{I}_t$ respectively, and solve for $\hat{Y}_t$, to

produce the IS curve:

$$\hat{Y}_t = -\frac{b}{1-\alpha}\hat{r}_t - \frac{\alpha}{1-\alpha}\hat{T}_t + \frac{\alpha}{1-\alpha}(\hat{G}_t + \hat{X}_t + \xi_t) \tag{6}$$

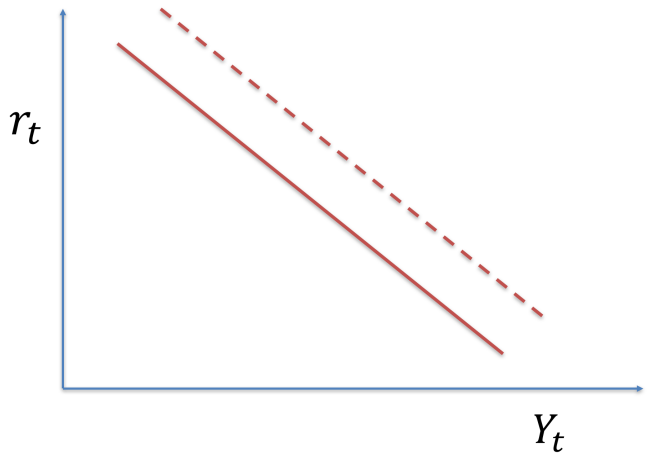


Figure 1: The IS Curve

where $\xi_t \equiv \xi_t^C + \xi_t^I$. The IS curve is shown in figure 1, i.e. the pair of output and interest rate implied by (6) holding fixed government spending and taxes, as well as the exogenous terms $\hat{X}_t$ and ξ_t. The dashed line shows the consequence of increasing government spending, cutting taxes or increasing $\hat{X}_t$ or ξ_t.

The IS curve is downward-sloping in figure 1 because lower interest rates increase spending. The reason for this is that a lower real interest rate increases investment demand. The higher investment demand in turn increases income, which triggers higher spending via consumption spending, which then triggers even higher income and thus even more consumption, and so on. This is the classic multiplier effect in the IS-LM model. In some variations of the IS-LM model the consumption function also depends on the real interest rate. This, however, does not change the resulting equation, just the interpretation of the parameter b.

3 The LM equation

Keynesians motivated the LM equation as follows: People hold money to facilitate transactions and as a form of asset. The more income they possess, the more money they tend to hold. In deciding their money balances, however, people consider the alternative use of the money, i.e. the opportunity cost of

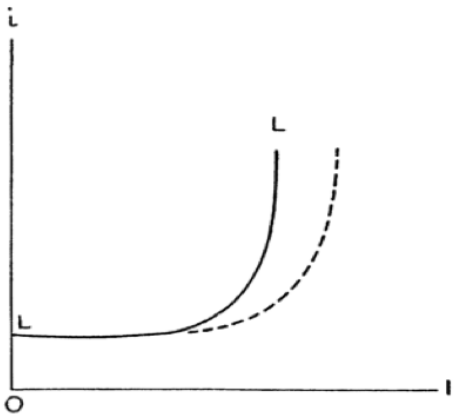


Figure 2: John Hicks's original diagram for the LM curve in his 1937 Econometrica article

holding money. The traditional measure of this opportunity cost is the short-term, risk-free nominal interest (even if it was often a subject of debate which interest rate was the relevant one). In its general form, a demand function for money can be written as

$$\frac{M_t}{P_t} = L(Y_t, i_t) \tag{7}$$

where $\frac{M_t}{P_t}$ represents the real money supply, where M_t is the quantity of money and P_t is the price level, i_t is the nominal interest rate. Typically, it is assumed that the derivative of the functions L are given by $L_y(.) > 0$ and $L_i(.) \leq 0$, as plotted in figure 2, taken from Hicks's original 1937 Econometrica article, where i denotes the interest rate and y output. It is the set of interest rate and output pairs consistent with some fixed money supply as given by equation 7. The dashed line shows the consequence of increasing the money supply. This LM curve is upward sloping in figure 2, as higher output is associated with more money demand. Hence, since the LM curve is plotted for a fixed M, the interest rate needs to increase as well.

As pointed out by Hicks, there will be a point at which the interest rate becomes so low that people will be indifferent between holding money and interest-bearing assets. This is what the modern literature calls the effective lower bound on interest rates (ELB). At the ELB money demand is "indeterminate." People are indifferent between holding money and bonds. Accordingly, theory cannot tell how much they will hold of each. This is an idea we will discuss in more detail later, in a more modern setting.

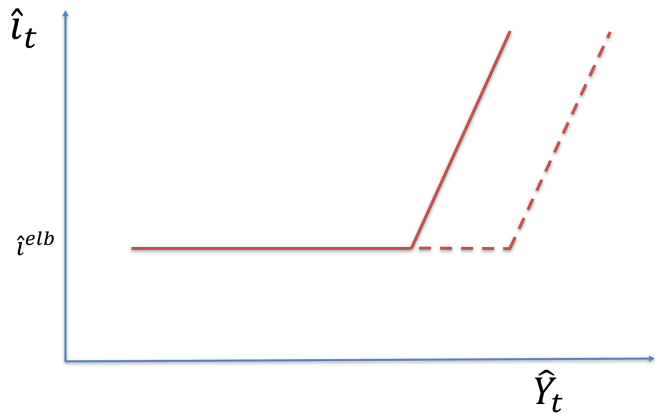


Figure 3: The LM Curve

To approximate the LM curve we consider the following simple representation of Hicks's figure:

$$\hat{M}_t - \hat{P}_t \geq c\hat{Y}_t - d\hat{\imath}_t \quad \hat{\imath}_t \geq \hat{\imath}^{elb} \tag{8}$$

where i^{elb} is the effective lower bound on the interest rate and at least one of the inequalities has to hold with equality at each time t. In other words, if the interest rate is above the ELB, the first equation holds with equality, while if the ELB is binding, the first inequality may apply. The coefficients c and d are greater than zero. This approximated LM curve is shown in figure 3, with the dashed line showing the effect of increasing the money supply.

4 The IS-LM Model and Equilibrium

Now that we have the two prerequisite equations, we can put them together to produce the unified IS-LM model. But first, we need to make a key assumption.

4.1 The Key Assumption of the IS-LM Diagram

Before combining the IS figure and the LM figure into one panel, observe a key difficulty in merging the two. They involve two different variables. While the IS curve depends on the real interest rate, r_t (i.e. how many units of real goods and services today you are willing to give up for one unit of goods and services tomorrow), the LM curve instead depends upon the nominal interest rate, i_t (how many dollars today you are willing to give up for one future dollar).

The real interest rate and the nominal interest are generally believed to be related to one another (an insight we will confirm later in a micro-founded model) via the Fisher equation. The Fisher equation is a basic non-arbitrage equation that has to hold for people to be indifferent between holding a bond that promises a nominal return and one that instead is indexed to inflation. The Fisher equation says that the real interest rate equals the nominal interest rate net of expected inflation

$$\hat{r}_t = \hat{\imath}_t - E_t \pi_{t+1} \tag{9}$$

where π_t denotes inflation, i.e. the percentage change in the price level.

In the first exposition of the IS-LM model, the key assumption is that prices are completely fixed, i.e.,

$$P_t = \bar{P} \tag{10}$$

The idea is that this is a reasonable approximation in the short run, even if in the long run it was acknowledged that price adjustment would take place. A famous justification for this leap is Keynes's observation that "in the long run, we are all dead."

Under this assumption, then

$$\hat{r}_t = \hat{\imath}_t \tag{11}$$

so that the IS and the LM curves can be plotted in the same diagram. As we will see, inflation expectations will play a fundamental role in later analysis. This leap, however, allows for a straightforward analysis of the IS-LM model with several important insights.

4.2 The Keynes and the Classics

The IS-LM curves are plotted together in figure 4. The figure shows two possible intersections. Hicks associated the intersection at the flat part of the LM curve with Keynes, and the one at the upward-sloping part of the LM curve with the Classics. According to Hicks, the two schools of thought could be understood by considering these two possibilities.

The Keynesian intersection, denoted by K, can be computed by combining the IS curve with the flat part of the LM curve, i.e. the case in which $\hat{\imath}_t = \hat{\imath}^{elb}$, while the Classics solution, denoted by C, can be obtained by combining the IS curve with $\hat{M}_t = c\hat{Y}_t - d\hat{\imath}_t$, as shown below:

$$\hat{Y}_t^K = -\frac{b}{1-\alpha}\hat{\imath}^{elb} - \frac{\alpha}{1-\alpha}\hat{T}_t + \frac{\alpha}{1-\alpha}(\hat{G}_t + \hat{X}_t + \xi_t) \tag{12}$$

$$Y_t^C = \frac{b}{(1-\alpha)d + cb}\hat{M}_t - \frac{\alpha}{1-\frac{bc}{d}}\hat{T}_t + \frac{\alpha}{1-\frac{bc}{d}}(\hat{G}_t + \hat{X}_t + \xi_t) \tag{13}$$

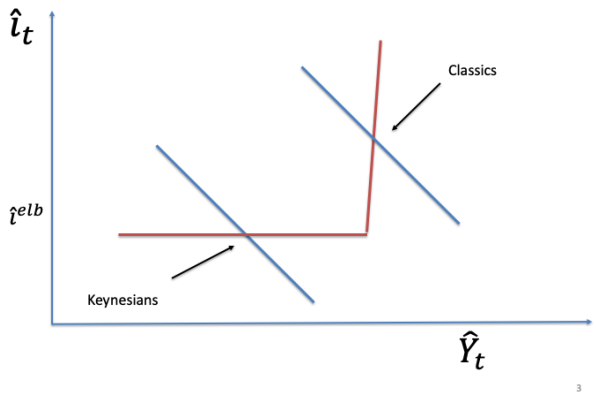


Figure 4: IS-LM diagram

Whether one assumes the intersection occurs at point K or point C has fundamental policy implications for monetary and fiscal policy. Consider first fiscal policy. An increase in $\hat{G}_t$ shifts out the IS curve as shown in figure 5. As shown in the figure, the effect is much larger if the economy is at K than at C.

An increase in $\hat{G}_t$ increases output by $\frac{\alpha}{1-\alpha}$ if the intersection is at K. For a value of $\alpha = 0.65$ this means that one dollar of government spending increases output by one dollar and 85 cents. When the intersection is C, however, the effect is $\frac{\alpha}{1-\frac{bc}{d}}$. A useful special case is when the LM curve is perfectly vertical, i.e. when d $\rightarrow 0$. In this case then $\frac{\alpha}{1-\frac{bc}{d}} \rightarrow 0$ and the increase in government spending has no effect on output. It only increases interest rate. The belief that d is small is a viewpoint Hicks attributes to the Classics. This special case is also useful in defining Monetarism, as discussed below.

While fiscal spending has a large effect at the intersection K, and little effect at C, the opposite is true for a monetary expansion. Figure 6 shows how a monetary expansion shifts out the LM curve. In the Keynesian case, as confirmed in equation 12, there is no impact on output. Meanwhile, according to the Classics perspective, output increases by $\frac{b}{(1-\alpha)d+cb}$.

The implication of which equilibrium reflects reality, C or K, thus has fundamental implications for policy. According to the Keynesian view fiscal policy was the main tool to battle a recession, while the Classical view was that monetary policy should be the main tool.

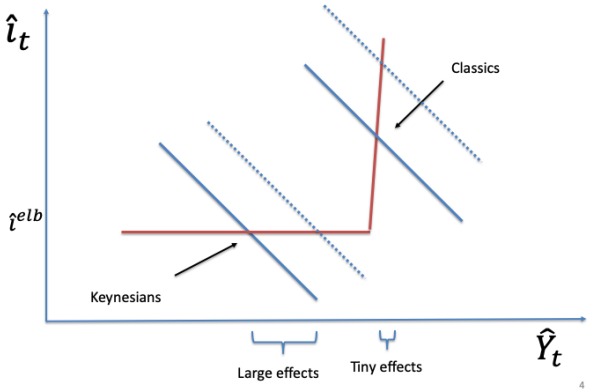


Figure 5: A Fiscal Expansion

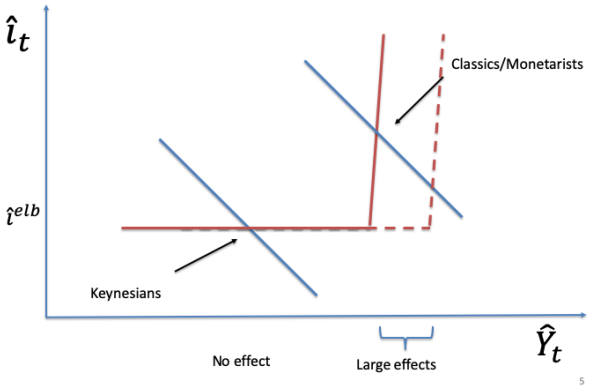


Figure 6: A Monetary Expansion

4.3 The Classics and the Monetarists

Following the Great Depression, the economic profession largely converged on the Keynesian view. Accordingly, the view was that fiscal policy was the most potent tool for fighting recessions. This, for example, was one reason for the tax cuts aimed at stimulating growth during the Kennedy administration. A major challenge to the Keynesian orthodoxy was issued by Friedman and Schwartz with the publication of the influential book *A Monetary History of the United States* in 1963, which we briefly discussed in the last chapter. The basic conclusion of that book was that all major contractions in US economic history were due to inappropriate monetary policy, in particular the Fed allowing the money supply to contract. Thus, the argument went, what was needed to cure recessions was not fiscal activism, but proper monetary policy that prevented abrupt contraction in the money supply. The basic viewpoint of Friedman and Schwartz became largely known as *monetarism*. While there is still debate on what, exactly, constitutes monetarism, a simple perspective on it is that it is a resurrection of the view Hicks attributed to the Classics, i.e., the perspective that the LM curve is approximately vertical (this viewpoint is also voiced by Friedman (1973) when Friedman attempted to formalize the analytical framework behind the his work with Schwartz). The case of the vertical LM curve corresponds to the case in which the velocity of money is constant. The velocity of money is defined as number of times each dollar in circulation is used to buy goods and services in a given time period (e.g. each year). Suppose d=0, and $\hat{\imath}_t > \hat{\imath}^{elb}$. Then, assuming the functional form (7) is linear at positive interest rates, the LM equations can be written as

$$M_t \frac{1}{c} = P_t Y_t \tag{14}$$

which can be recognized as the fundamental equation underpinning the Quantity Theory of Money, where $\frac{1}{c}$ is velocity of money. The key assumption behind the quantitative theory of money is that velocity is approximately constant, both at high and low interest rates. Thus, Friedman came down firmly with the Classics, and the intersection C provides a useful way to think about Monetarism. But who was right? The Keynesians or the Classics/Monetarists?

4.4 The Keynes and the Classics: Resolving the Debate

A natural question to ask is whether or not this controversy can even be resolved. In the context of the IS-LM model, I think the answer is no. The problem is that the IS-LM model is static. Hence, there is no way of discriminating between long-term and short-term interest rates, nor is there a distinction between risk-free and risky assets. It is not even clear how to start addressing these types

of questions inside the model. For example, monetarists long argued that the y-axis of the IS-LM model should refer to long-term risky nominal rates, all of which remained positive during the Great Depression in the US (and the Great Recession). Thus, they argued that the effective lower bound had not been reached. Meanwhile, some Keynesians argued that the correct variable should be the risk-free nominal interest rate, which was close to zero. Evidently it could not go much below that, for nobody would lend one dollar unless he is guaranteed at least one dollar back. One of the key advantages of the modern approach to macroeconomics is that there will be a straightforward way of refereeing debates over what type of variables are relevant for the analysis. Because we will derive the models from the basic decision foundations of each agent, a powerful approach will be to simply ask the agents inside the model what variables are relevant for their decisions. One application of this, which we will return to, is to referee the debate between the Keynesians and the Classics/Monetarists. As we will see, either one of them can be shown to be right or wrong, depending on the assumed policy regimes. But all of that will have to wait until later.